LGCNet: A Cloud Detection Method in Remote Sensing Images Using Local and Global Semantics

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***Abstract*— Detecting and eliminating clouds is a crucial step in remote sensing image (RSI) preprocessing. The removal of clouds can significantly enhance the performance of subsequent remote sensing applications. Existing deep learning (DL)-based cloud detection methods extract semantic information to improve feature representation and, subsequently, detection performance. However, these methods do not fully utilize the potential of context semantic information. Besides, to capture semantics from large receptive fields, they employ convolution operators with large kernel sizes, which results in high computational costs. Thus, these computationally heavy models are not suitable for resource-limited devices, particularly satellites. To address this issue, we propose a cloud detection model, LGCNet. This model efficiently extracts both local and global contextual information, fully utilizing semantics while reducing resource usage. LGCNet is built on an encoder–decoder structure. Specifically, the encoder extracts local scale-aware semantics through proposed local semantic blocks (LSBs), which are then skip-connected to the decoder. This approach provides adaptive and diverse local contextual information. On the top of the encoder, the high-level global semantics are captured via the proposed global feature TransBlock (GFTB). A variety of extracted semantics ensure improved detection performance. We evaluate the proposed method using two public datasets: LandSat8 and Moderate-Resolution Imaging Spectroradiometer (MODIS). We conducted experiments on both a server and an edge computing device. Our extensive experiments revealed that LGCNet outperforms other lightweight cloud detection and semantic segmentation methods in terms of performance and computational load.**

***Index Terms*— Cloud detection on remote sensing images (RSIs), lightweight network, semantic information extraction, transformer.**

I. Introduction

With the rapid development of remote sensing technology, this field has attracted a growing number of researchers. Applications of remote sensing are built upon remote sensing images (RSIs) and span various areas, including image classification [1], disaster risk management [2], vegetation engineering [3], building extraction [4], object change detection [5], and object detection [6], [7], among others. However, a significant challenge in utilizing RSIs is that nearly 60% of the Earth's surface is obscured by clouds [8]. This inevitable cloud occlusion heavily impedes the quality and effectiveness of applications relying on RSIs. Therefore, accurately identifying areas covered by clouds becomes an essential step in the preprocessing of RSIs.

With the development of aerospace and remote sensing technology, the importance of onboard cloud detection is increasing. One of the most significant benefits of onboard cloud detection is the conservation of downlink bandwidth between satellites and Earth [9]. For instance, on satellites, if the area covered by clouds in an RSI exceeds a predetermined threshold, the image is considered invalid for subsequent RSI applications and therefore not transmitted to Earth. Furthermore, RSIs with cloud will not compromise the performance of subsequent onboard remote sensing applications. In the past, cloud detection models were trained and tested on servers using transmitted RSIs. However, with the advent of modular satellites equipped with powerful computational processors such as GPUs, onboard artificial intelligence (AI) inference has become a reality in recent years [10]. As such, there is a compelling need to develop a lightweight cloud detection method that delivers superior performance while consuming minimal resources.

Cloud detection methods are typically grouped into three distinct categories: threshold-based methods, machine-learning-based methods, and deep-learning-based methods. Threshold methods utilize empirical spectral and wavelength-specific thresholds to distinguish clouds from non-cloud areas at a pixel level [11]. While these methods are relatively straightforward, they lack the necessary robustness to manage the various environments found in RSIs. Moreover, when applied to hyperspectral RSIs, the computational burden can become significantly high due to complex threshold calculations. Machine learning methods, on the other hand, employ additional physical attributes such as texture and geometry. These attributes are first processed by domain experts and then used by statistical learning algorithms, such as support vector machines (SVMs) [12] and random forest (RF) [13]. However, the performance of cloud detection in this process heavily depends on the manually crafted input features, which can be easily influenced by other factors. As a result, the performance of machine learning methods is often unsatisfactory.

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Recently, the mainstream of cloud detection is methods based on deep learning (DL). The power of DL, with its inherent ability to automatically extract discriminative features, has yielded extraordinary results in both computer vision and natural language processing (NLP). Many researchers have proposed cloud detection models based on convolutional neural networks (CNNs) [14], and these models have made a huge step forward in cloud detection. In CNNs, convolution operators are capable of extracting spatial and semantic features. To enhance the performance, many models stack more modules based on convolution operators and use larger kernel sizes to capture semantics within larger receptive fields. However, these methods significantly increase the computational load. Despite their high-accuracy detection results, the extensive resource consumption inhibits their application in on-orbit cloud detection tasks. With the advancement of aerospace technology, efficient cloud detection has become a trending research area. Some researchers have attempted to minimize resource consumption by employing fewer layers and more lightweight convolution operators [10], but these modifications have led to a substantial decline in detection performance.

Using semantic segmentation models trained on natural scene images (NSIs) for pixel-level classification can be an effective strategy for cloud detection. Moreover, efficient segmentation models designed for mobile devices have demonstrated outstanding performance [15]. However, the transition from NSIs to RSIs is not straightforward and presents unique challenges for cloud detection tasks. First, cloud deformation on appearance is nonrigid, and the shape and thickness of clouds can vary greatly. Second, in RSIs, thin cloud is often transparent, and their spectral feature expression is partially impacted by their underlying surface. It makes the feature expression of the thin cloud different from that of the normal cloud. Therefore, aligning features of normal clouds and thin clouds is difficult but important. Third, cloud and cloud-like objects, such as snow and ice, have similar feature expressions, adding further complexity to the challenge of distinguishing clouds.

In our cloud detection research, we found that including more contextual information can reduce the number of false detections and missed detections caused by the unique characteristics of RSIs. In scenario 1, as shown in Fig. 1(a), the blurred boundary of the cloud, primarily covered by thin clouds, contains features of the surface underneath. Fortunately, the pixels at this boundary are adjacent to thick clouds, and this contextual information can be useful for detecting thin clouds. In scenario 2, the presence of clouds alongside snow or ice also elevates the misclassification rate, as shown in Fig. 1(b). Fortunately, snow and ice exist in specific scenarios, e.g., snow covered on the top mountain appears in mountain topography, and ice sheet covers the ocean normally in polar regions. By capturing the local and global semantic information, detection accuracy can be increased.

To extract semantics as much as possible, previous methods utilize normal convolution with a large kernel size [16], [17]. However, the resulting receptive field remains local. Although this limitation can be somewhat mitigated by increasing the kernel size, this approach introduces a larger number of parameter calculations. Furthermore, the complex shape of the cloud, especially in boundary areas, presents additional challenges [18]. Standard convolution is unable to adjust its receptive fields to capture semantics effectively in these irregular and nonuniform areas. As a result, only a limited amount of meaningful context can be gathered through standard convolution. To obtain multiscale semantics, the typical approach has been to stack atrous convolutions with varying dilation rates [19]. However, this method introduces a "gridding issue" that disrupts feature continuity and hampers improvement in cloud detection performance [20].

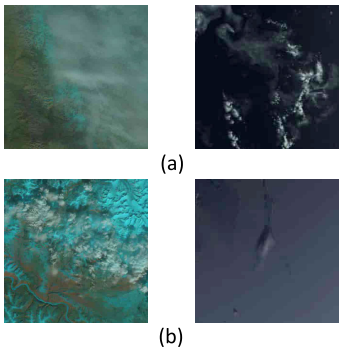


Fig. 1. Samples of RSIs. (a) Blurred cloud boundary. (b) Coexistence of clouds and snow or clouds and ice.

Motivated by this, we propose a network to efficiently address cloud detection through the extraction of local and global contextual information (LGCNet). The network is based on an encoder–decoder architecture. In the encoder, the feature maps are encoded by the proposed feature encoding module. Feature encoding block (FEB) can extract the spatial features on feature maps of different resolutions. Moreover, the encoder captures high-level global semantics via the proposed global feature TransBlock (GFTB). GFTB obtains global attention by extending the transformer block. To extract local semantics effectively, we design a local semantic block (LSB). LSB incorporates a lightweight deformable convolution operator and an extended window transformer. These two types of local semantics, capable of dealing with a variety of cloud shapes, are then fused within the LSB. Then, these local semantics are connected to the decoder and fused with the corresponding feature maps in the decoder through feature decoder module (FDM). By effectively capturing local and global semantics effectively, LGCNet minimizes the incidence of cloud misclassification. Furthermore, LGCNet is computationally efficient. We have extended the deformable convolution operator into a lightweight version and have reduced the computational load of the transformer by restricting the size of the window used. The main contributions of our work include the following.

1) We propose the LSB that fuses inside window context and context in the adjusted local receptive field to capture multiscale semantics information in cloud detection.
2) We propose the GFTB to capture the high-level long-range scene semantics to boost effective cloud detection.

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3) We design the lightweight cloud detection model LGCNet built on feature encoding module, LSB, and GFTB. Extensive experiments on two open datasets, LandSat8 and Moderate-Resolution Imaging Spectroradiometer (MODIS), present that LGCNet achieves the state-of-the-art performance. Besides, from experiments on the device NVIDIA Jetson Xavier NX, the efficiency of LGCNet is satisfied in the edge computing scenario.

We organize the rest of this article as follows. In Section II, related works on cloud detection are briefly reviewed. In Section III, we introduce the architecture of LGCNet in detail. In Section IV, experimental results on LandSat8 and MODIS are presented. Besides, ablation studies on the proposed modules are conducted. Finally, we conclude this work in Section V.

II. Related Work

A. Cloud Detection Methods

As a traditional cloud detection branch, threshold-based methods usually adopt the clouds' physical characteristics to generate cloud masks. The criteria, which are based on the empirical threshold, discriminate clouds from their underlying surfaces [21]. For example, Fmask exploits the Landsat top-of-atmosphere (TOA) reflectance and brightness temperature (BT) of images to generate the threshold for cloud segmentation [11]. In Tmask [22], extra-temporal information is further captured from the noncloud pixels detected by Fmask to boost the detection accuracy. Based on Fmask, MFmask [23] developed a new algorithm to automatically detect cloud and cloud shadow. By adding the digital elevation models, water and land pixels can be better separated from clouds in mountainous areas. The performance of threshold-based methods is limited by only depending on physical characteristics and empirical thresholds. Other features, such as luminance and texture, are barely used. As another traditional branch, machine learning-based approaches, such as the SVM [24], Markov random fields [25], and RF [13], take various manual-crafted features as input and analyze typical spectral, textural, and spatial features of clouds. However, the discriminable manual-crafted features are difficult to obtain in complex scenarios.

Recently, with the neural network methods' great achievement in computer vision, this innovative technology has also been widely adopted in remote sensing tasks. In [26], a dilated network is used to handle varying patch sizes to improve pixel-wise classification accuracy in RSIs. The proposed model can capture multicontext characteristics from diverse contexts. In the task of semantic segmentation on high-resolution RSIs, Jin et al. [27] leverage spatial information from edge detection and balance detail and semantic information between high- and low-level features. Recently, to fuse features from multiple data sources, a multiscale interactive information extraction method is used to extract multiscale information [28]. Wang et al. [29] design a multistage self-guided separation network for RSI scene classification. It aims to address the challenges of insufficient feature differences in interclass samples and inconsistent feature representation in intraclass samples.

Considering cloud detection, CNNs are first adopted, and they extract features automatically and boost the accuracy of classification tasks with captured expressive features [30], [31]. In [14], a CNN-based network is designed on an encoder–decoder structure. It generates the cloud masks with RSI thumbnails, which are of low resolution and have less spectrum information. Guo et al. [32] target specifically on cloud-snow coexistence scenarios. In this model, the specific channel and spatial attention fusion models are designed to tackle such complex scenarios. Wu et al. [33] propose the boundary nets, which contains a scalable-boundary net and a differentiable-boundary net. It generates and fuses multiscale cloud masks to tackle the problem of variable visual forms of cloud. Yin et al. [34] improve the cloud detection accuracy by deeply extracting features of cloud and snow.

B. Lightweight Models

Considering that the hardware devices technology is evolving rapidly, the models that can be deployed on edge computing devices are in urgent need. Therefore, more researchers have focused on the lightweight models for semantic segmentation task and gained remarkable tradeoffs between efficiency and effectiveness. Since semantic segmentation is also a pixel-level classification task [35], the used methods are worthy to be reviewed.

The most used method to reduce computation complexity is adopting the separable convolution [36] for feature extraction. The separable convolution consists of a depthwise convolution and a pointwise convolution. The depthwise convolution conducts convolution operations on the channel level, and the pointwise convolution is responsible for fusing the extracted channel-level features. Therefore, the number of parameters can be reduced significantly. In ShuffleNet [37] and Deeplab V3+ [38], the separable convolution is adopted in blocks to reduce calculation complexity. Facilitating feature reuse [39] and handling spatial details and semantics information separately for a few used parameters [40] are other widely utilized methods in efficient models.

In the semantic segmentation task, many models improve the segmentation performance by increasing the receptive field in semantics capture. The atrous convolution can provide a large receptive field by setting a large dilation rate [38]. Also, the atrous convolution has the same number of parameters as the normal convolution. Multiscale semantic information helps in performance [41], [42]. Yang et al. [43] show the capability of increasing the segmentation accuracy by capturing multiscale semantic information. However, the "gridding issue" [20] is inevitable in the dilated convolution, which will cause miss or false segmentation in results.

Recently, as a new type of neural network, transformer gains significant success in NLP tasks. Then, it is soon ported to the computer vision research area. Different from CNNs, the transformer utilizes the attention mechanism to explore the relations between features for final classification [44]. In Swin Transformer [45], input images are divided into patches, and the attention of patches is calculated inside the defined local window. Features across windows are extracted via the shifted

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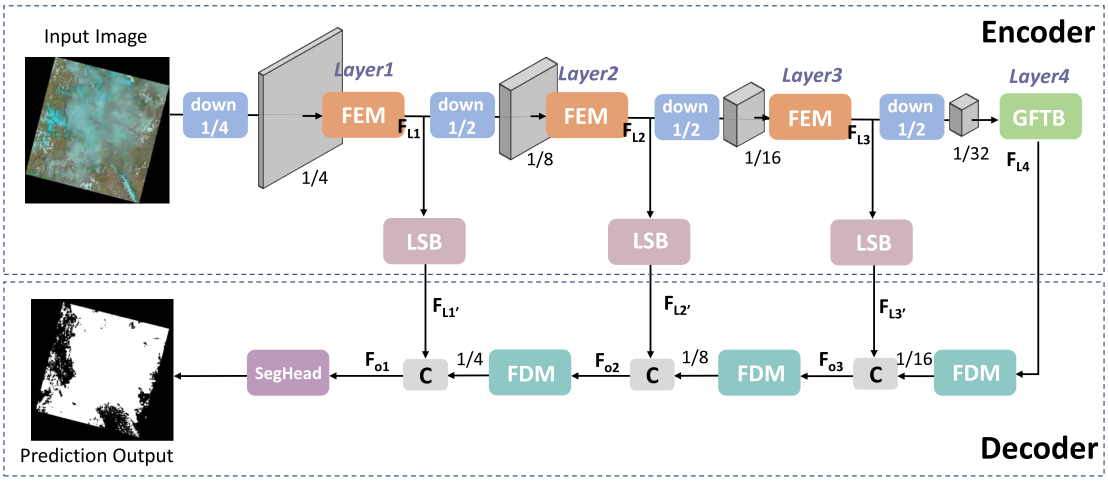


Fig. 2. Architecture of the proposed LGCNet. The encoder consists of the backbone built on the proposed FEM. LSB extracts divers local semantics to boost cloud detection. GFTB targets the high-level global semantics capture. In the decoder, the extracted semantics are compensated to feature maps, and FDM fuses these features.

window partitioning. However, Swin Transformer still has heavy computation complexity. TopFormer [46] takes tokens from various scales to generate scale-aware semantic features. The computation complexity is reduced by adopting a transformer block on feature maps with a low resolution. However, the model's backbone contains stacked blocks and there still exists room for further complexity reduction. Besides, to further boost detection accuracy, specific designs against cloud characteristics are required.

With the trend of designing onboard models for satellites, researchers have studied much about efficient cloud detection methods. Li et al. [47] applied a lightweight CNN for cloud detection on Sentinel-2A images. To reduce the number of parameters, the model takes depthwise separable convolution and shared kernel between channels. However, depthwise separable convolution has fixed local receptive fields and kernel sharing maintains the computation complexity. The LWCDnet [19] is designed in encoder–decoder architecture and utilizes lightweight operators for feature extraction. Even though the model resource is saved, the accuracy of cloud detection is not satisfactory. In [18], the receptive fields are adaptive adjustments by using the deformable convolution. However, calculating the extra offset in deformable convolution is computation-consuming. Consequently, an efficient cloud detection method with high accuracy is in urgent demand.

III. Method

A. Overall Framework

Fig. 2 presents the architecture of the proposed cloud detection model: LGCNet. It is built on the encoder–decoder architecture. In the encoder, multiscale cloud features are extracted gradually via the proposed feature encoder module (FEM). On the last layer of the encoder, the designed module GFTB enhances the expression of feature maps by adding global semantics. To augment the representation, rich local semantics are obtained via the proposed LSB and skip connected to the decoder. In the decoder, the corresponding feature maps are fused in FDM and recovered to the same size as the input image gradually.

Specifically, multichannel RSI I_I is first downsampled to $1/4 \times 1/4$ size of the input image. Then, the FEM extracts mainly spatial information from feature maps. The output of Layer1 can be denoted as

$$F_{L1} = \mathrm{FEM}\left(I_{(1/4)}\right)\left(\mathrm{down}_{1/4}(I)\right) \tag{1}$$

where $I_{1/4}$ denotes the feature maps that are of $1/4$ size and down represents the downsampling operator, which reduces the size of feature maps and meanwhile increases the number of feature map channels. The operations are the same in Layer2 and Layer3

$$F_{L2} = \mathrm{FEM}\left(I_{(1/8)}\right)\left(\mathrm{down}_{1/2}(F_{L1})\right) \tag{2}$$

$$F_{L3} = \mathrm{FEM}\left(I_{(1/16)}\right)\left(\mathrm{down}_{1/2}(F_{L2})\right) \tag{3}$$

where F_{L1}–F_{L3} represent the output feature map of Layer1, Layer2, and Layer3, respectively. The output of the encoder can be stated as

$$F_{L4} = \mathrm{GFTB}\left(I_{(1/32)}\right)\left(\mathrm{down}_{1/2}(F_{L3})\right). \tag{4}$$

Here, the designed block GFTB encodes high-level global semantics to boost feature expression.

In the decoder, feature maps are reverted gradually. Outputs of Layer1, Layer2, and Layer3 are added with local semantics captured by LSB and then compensated in decoder

$$F_{L3'} = \mathrm{LSB}(F_{L3}) \tag{5}$$

$$F_{O3} = \mathrm{concat}(F_{L3'},\ \mathrm{FDM}(F_{L4}) \tag{6}$$

$$F_{L2'} = \mathrm{LSB}(F_{L2}) \tag{7}$$

$$F_{O2} = \mathrm{concat}\left(F_{L2'},\ \mathrm{FDM}\left(I_{(1/16)}\right)\right) \tag{8}$$

$$F_{L1'} = \mathrm{LSB}(F_{L1}) \tag{9}$$

$$F_{O1} = \mathrm{concat}\left(F_{L1'},\ \mathrm{FDM}\left(I_{(1/8)}\right)\right). \tag{10}$$

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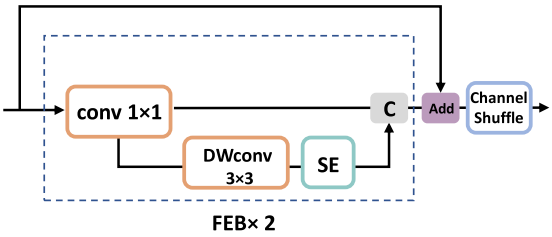


Fig. 3. Proposed FEM. It is designed to extract expressive features with low resource consumption. In FEM, the FEB is stacked twice and added with the residual input feature maps.

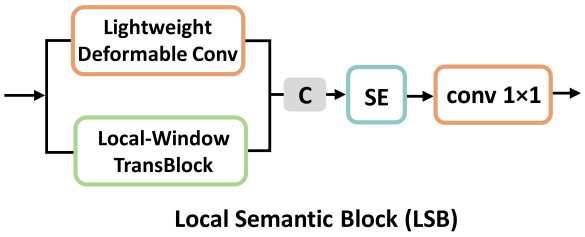


Fig. 4. Proposed LSB. It can capture two types of local semantics via a lightweight deformable convolution and a proposed local-window transformer block.

Here, the FDM upsamples the feature maps to two times larger and fuses feature maps from the encoder and decoder effectively. In addition, FDM consists of a normal convolution, a batch norm, and a Hardswish active function

$$F_{\mathrm{FDM}} == \mathrm{Hardswish}\left(\mathrm{BathNorm}\left(\mathrm{conv}\left(\underset{(1/x)}{I}\right)\right)\right). \quad (11)$$

Finally, the cloud mask is generated and restored to the original resolution via the SegHead

$$F_{\mathrm{Mask}} = \mathrm{SegHead}(F_{O1}). \quad (12)$$

Here, the SegHead is built on three consecutive convolution operators and a PixelShuffle operation

$$F_{\mathrm{SegHead}} = \mathrm{PixelShuffle}(\mathrm{conv}(\mathrm{conv}(\mathrm{conv}(I)))). \quad (13)$$

In the training phase, the cross entropy is adopted as the loss function to calculate the loss between the generated cloud mask and the ground truth

$$L_{\mathrm{CrossEntropy}} = -\sum_{c=1}^{C}\left(p_c \times \mathrm{logq}_c\right) \quad (14)$$

where p_c denotes the reference cloud mask and q_c is the inference mask. C represents the classes, where C equals 2: cloud and noncloud.

B. Feature Encoder Module

The low-level features contain rich detailed information. Convolution operators are normally adopted to capture such features. Previous works achieve high performance by stacking modules built on convolutions. However, it has been proven in [48] that redundancy existing in feature maps can guarantee a comprehensive understanding of input features. To compact the models, cheap operators are deployed to generate the ghost feature maps separately to enhance the feature expression with low resource consumption. In FEM, we follow the GhostNet and generate ghost features (Fig. 3). By observing that the separate convolution gains comparable performance but needs less computation, the depthwise convolution is utilized in FEM. Besides, an SE-Block [49] emphasizes the channel relationships in ghost features and attracts more attention to useful ones. In FEM, the proposed feature encoder block is stacked twice and added with the residual input feature maps to avoid the vanishing gradient problem [50]. Finally, a shuffle block [37] fuses group isolated features from depthwise convolution, ghost features, and the features from the residual connection efficiently. Therefore, FEM can extract low-level features completely with a low calculation complexity.

C. Local Semantic Block

In previous works, the large receptive fields are achieved by adopting convolutions with large size kernel [51] or using the atrous convolutions with large dilate rate [43]. However, the fixed geometric structures of convolution operators limit the captured receptive fields. Also, for each convolution unit, it can only sample the input feature map at fixed locations [52]. Besides, the shape of clouds varies, and it further weakens the convolutions' ability to capture useful semantics. As shown in Fig. 4, we design the LSB to extract two types of cloud context information.

LSB consists of two blocks, a lightweight deformable convolution and a proposed local-window transformer block. The deformable convolution can effectively capture the clouds' semantics within the cloud boundary area. As stated, the cloud boundary normally has various geometric structures [18]. As shown in Fig. 5, a 2-D direction offset is added to the regular grid in deformable convolution, and therefore, the sampling is at an offset location.

$$y(p) = \sum_{p\in\Re} w(p)\cdot x(p_0 + p + \triangle p) \quad (15)$$

where p enumerates the locations in R, and we take $R = \{(-1,-1),(-1,0),\ldots,(0,1),(1,1)\}$ in the 3×3 kernel. For each location p_0 on the output feature map, the deformed location is $p+\triangle p$. Such arbitrary deformation can adapt to cloud geometric forms and obtain an adaptive receptive field. Besides, we adopt the separable convolution in calculating the offsets to gain a low computation complexity.

The proposed local-window transformer block utilizes the advantages provided by the transformer and gains intensive attention-based semantics. Different from convolution operators, the transformer has the capability for gathering generalized global context [44]. As shown in Fig. 6, for the pixel colored red, the receptive field of convolution with 3×3 kernel size and dilate rate that equals 1 is colored yellow and dilate rate that equals 2 is colored blue. In the 5×5 window, the combination of convolutions cannot cover the intensive semantics. Besides, such a combination introduces much more computation. In contrast, the transformer can include inside window pixels and calculate the attention-based contextual information. Furthermore, the multiscale semantics can be

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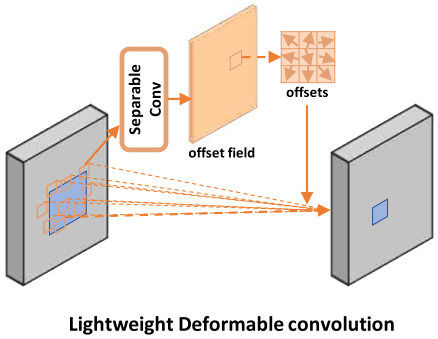


Fig. 5. Lightweight deformable convolution. Here, an offset is added to adapt to arbitrary cloud forms for capturing the adaptive receptive field.

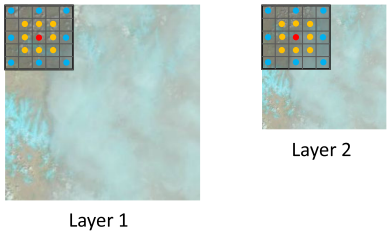


Fig. 6. Comparison of receptive fields by the convolution operator and the transformer. The dots represent the pixels in receptive fields of the convolution operator (yellow: dilate rate = 1 and blue: dilate rate = 2). The square area represents the pixels in the receptive fields of the transformer.

extracted by adopting the same transformer window size on hierarchically stacked layers. Feature maps from Layer1 and Layer2 have resolutions of $H/4 \times W/4$ and $H/8 \times W/8$, respectively; when we take window size equals 12, the receptive fields of each layer contain 48×48 and 96×96. Since the attention calculation is limited inside the window, the computation burden is very little.

Specifically, the local-window transformer block takes feature map $F^{C\times H\times W}$ as input. To calculate the inside window semantics, the feature map is arranged to $F^{H\times W\times C}$ and partitioned into windows $F^{(h\times w)\times(p\times p)\times C}$, where $p \times p$ is the window size and $h \times w$ presents the number of divided-out windows. Then, the multihead attentions are calculated inside windows to obtain local semantics. For each head, the calculation is formulated as

$$H_i = \max\left[\frac{W_i^Q M_p^Q \times W_i^K M_p^K}{\text{scale}}\right] \times W_i^V M_p^V \qquad (16)$$

where M_p^Q, M_p^K, and M_p^V denote the query, key, and value for multihead attention, respectively; and scale means the scale factor. Then, the feature map is to rearrange to input shape $F^{C\times H\times W}$ (Fig. 7).

Then, the local adaptive and dense semantics are combined and fused together. They boost the cloud detection accuracy effectively.

D. Global Feature TransBlock

Besides the local semantics, global semantics are also critical for cloud detection. In CNNs, the global context information is extracted via pooling operators [53]. However, insufficient extraction normally introduces miss detection and

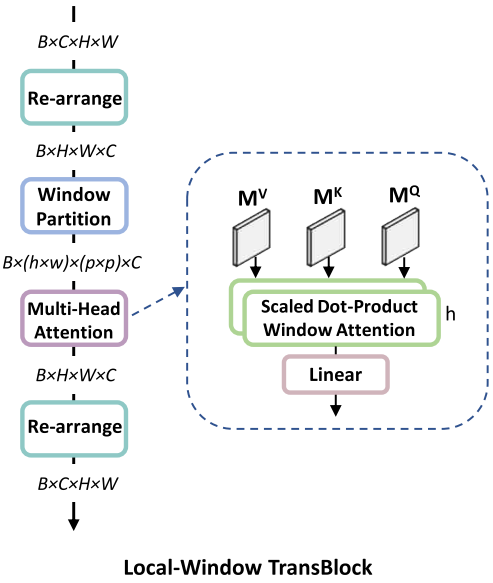


Fig. 7. Proposed local-window TransBlock. Feature maps are partitioned in windows and the semantics are calculated in window.

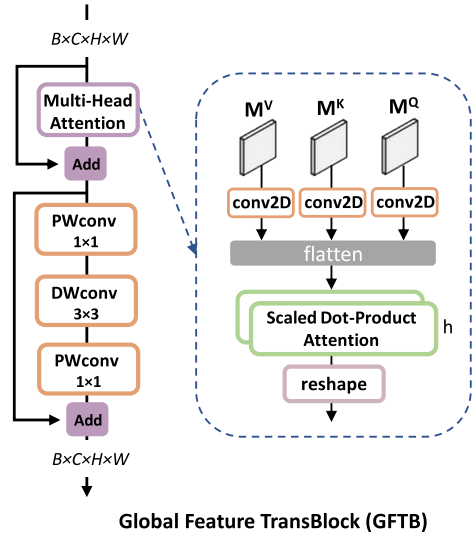


Fig. 8. Proposed GFTB. High-level semantics are calculated for global semantic extraction.

false detection [54]. Differently, the transformer has the ability to obtain the generalized global context and meanwhile focus on important areas. Since the mechanism of the transformer is self-attention, the correlation between pixels is calculated and the participation degree of key-area pixels is higher [55]. Therefore, we design the GFTB on the top level of the encoder for global semantic extraction. Since the feature map resolution of the top level is $1/32 \times 1/32$, the calculation complexity is limited to low.

As shown in Fig. 8, taken the feature map $F^{C\times H\times W}$ as input, in the multihead attention operation, the query, key, and value M_p^Q, M_p^K, and M_p^V are processed first via 2-D convolutions. Then, the multihead pixel-level attention flattens features are calculated on the flattened features. In the end, the features are reshaped back to a feature map of size $F^{C\times H\times W}$. Following the

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TABLE I
STATISTICS OF LANDSAT8 AND MODIS DATASETS

	LandSat8	MODIS
♯ Bands	4	10
♯ Training	7000	17880
♯ Valid	1400	1200
♯ Test	9201	2250
Patch Size	384×384	512×512

attention part, lightweight convolution operators are stacked to further boost the global semantics feature expression.

IV. DATASET AND EVALUATION METRICS

A. Datasets

To evaluate the cloud detection performance of the proposed LGCNet, we use two public datasets: LandSat8 and MODIS (Table I). LandSat8 is released in [56] and is widely used in cloud detection model validation. It contains 18 LandSat8 images of size 1000×1000 for training and 20 same-size images for testing. We cropped these images into patches with the size 384×384. In experiments, the training set contains 8400 patches, and the testing set contains 9201 patches. For each patch, the multispectral image is of four channels: red (band 4), green (band 3), blue (band 2), and near infrared (band 5). To assess the effectiveness of semantics extracted from LGCNet in a thin-cloud context, we have selected 484 patches from LandSat8 to create a sub-dataset specifically for this test. All the selected patches are predominantly covered by thin clouds.

MODIS is an Earth observation system in NASA [57]. The images in MODIS dataset are from the MODIS Level 1B product. MODIS dataset gathers images in the first five days of each month in 2005. After removing the problematic images, there are finally 1422 RSIs. Each of them contains ten informative channels (bands 1, 3, 4, 18, 20, 23, 28, 29, 31, and 32) of the original 36 bands. These images are divided into a training set (1192 samples), a validation set (80 samples), and a test set (150 samples). After cropping images into overlapping patches with size 512×512, the training, validation, and test sets contain 17 880, 1200, and 2250 patches, respectively.

B. Evaluation Metrics

To evaluate the method performance, we include the widely used metrics: Jaccard index, precision, recall, mean intersection over union (MIoU), F1-Score, and overall accuracy

$$\text{JaccardIndex} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{17}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

$$\text{OverallAccuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{21}$$

$$\text{MIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{p_{ii}}{\sum_{i=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}} \tag{22}$$

where TP, TN, FP, and FN denote the cloud correct prediction, noncloud correct prediction, false-positive incorrect detection, and false-negative incorrect detection, respectively; and MIoU is widely adopted to combine the accuracy of both cloud pixels and noncloud pixels in the calculation.

For efficiency measurements, we collect the related metrics, including giga floating-point operations per second (GFLOPs), which denote the computation complexity, the number of model parameters, time spent for inference per image, model size, and memory occupied by the model.

Specifically, we conduct experiments on the edge computing device: NVIDIA Jetson Xavier NX. NVIDIA Jetson Xavier NX is an advanced embedded AI supercomputer. It is equipped with a GPU of 384-core NVIDIA Volta @ 1100 MHz with 48 Tensor Cores. It can maximally deliver up to 21 TOPS of compute under 15 W of power [66]. By conducting experiments on the device side, we collect the inference time and power usage of models.

C. Implementation Details

In this study, all networks are all trained with the Pytorch framework on Ubuntu 16.05 with two RTX 3090 GPUs and optimized by the Adam optimizer. The learning rate starts from 0.01 and decays with the policy that from epoch 36, the learning rate decays to 0.008; from epoch 65, the learning rate decays to 0.005; and from epoch 85, the learning rate decays to 0.003. Training runs in 100 epochs. All methods are trained with the same configuration and settings without pretrain.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Baseline Methods

To compare the proposed LGCNet with the state-of-the-art methods, we take the baseline methods from cloud detection models (U-Net [58], Deeplab [38], CS-CNN [59], RS-Net [60], MF-CNN [61], MFGNet [62], CDFM [47], and DAB-Net [18]) and lightweight semantic segmentation methods (MobileNet [15], ICNet [64], DFAnet [35], LEDNet [65], BiseNet-V2 [40], and TopFormer [46]).

B. Cloud Detection Results on LandSat8 Dataset

In Table II, the experimental results of state-of-the-art cloud detection and lightweight semantic segmentation methods on the LandSat8 dataset are summarized. From the results, our proposed LGCNet outperforms all the other methods in all metrics except recall. CDFM [47] and Bisenet-V2 [40] obtain a marginally better recall score than our model. LGCNet utilizes the lightweight operator: depthwise separable convolution, in its FEM, LSB, and GFTB modules. This lightweight operation reduces computational demands, but some pixels may be omitted in feature encoding at the pixel level. This omission can lead to an increase in false-positive detection,

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TABLE II

COMPARISON WITH CLOUD DETECTION METHODS AND SEMANTIC SEGMENTATION METHODS ON THE LANDSAT8 DATASET. THE LAST LINE SHOWS THE DIFFERENCE BETWEEN OUR LGCNET AND THE HIGHEST VALUE OF OTHER METHODS

Category	Methods	Jaccard	Precision	Recall	MIoU	Overall	F1-Score
Cloud Detection Methods	Fmask[11]	68.4	55.2	68.8	47.9	51.7	61.3
	U-Net[58]	85.0	93.2	90.6	88.9	94.9	91.9
	CS-CNN [59]	73.6	87.9	91.2	81.0	95.3	89.5
	RS-Net[60]	85.8	93.4	91.4	89.5	95.2	92.4
	MF-CNN[61]	87.0	**95.1**	91.0	90.4	95.6	93.0
	MFGNet[62]	85.2	93.3	95.6	89.3	95.3	94.4
	CDFM[47]	87.4	94.4	96.2	91.0	96.0	95.3
	DABNet[18]	86.2	94.1	95.3	90.0	95.5	94.7
	BABFNet [63]	85.5	94.4	94.2	89.3	95.1	94.3
Semantic Segmentation Methods	MobileNet[15]	85.5	93.5	95.4	89.5	95.3	94.4
	ICNet[64]	85.4	93.0	95.5	89.4	95.3	94.2
	DFAnet[35]	85.4	92.1	93.1	86.3	93.7	92.6
	LEDNet[65]	85.4	93.5	95.4	89.5	95.3	94.4
	BiseNet-V2[40]	87.9	94.6	**96.4**	91.3	96.0	95.5
	TopFormer[46]	84.4	93.5	94.4	88.7	94.9	93.9
	LGCNet	**88.4**	**95.1**	96.1	**91.6**	**96.3**	**95.6**
	+/− diff	+0.5	+0.0	-0.3	+0.3	+0.3	+0.1

TABLE III

COMPARISON WITH CLOUD DETECTION METHODS AND SEMANTIC SEGMENTATION METHODS ON THIN-CLOUD DATASET. THE LAST LINE SHOWS THE DIFFERENCE BETWEEN OUR LGCNET AND THE HIGHEST VALUE OF OTHER METHODS

Category	Methods	Jaccard	Precision	Recall	MIoU	Overall	F1-Score
Cloud Detection Methods	U-Net[58]	71.0	77.4	82.1	64.2	80.1	79.7
	CS-CNN [59]	61.9	78.8	76.0	58.3	73.9	77.4
	RS-Net[60]	70.0	78.2	81.4	64.4	79.0	79.8
	MF-CNN[61]	72.8	79.6	82.8	66.9	80.9	81.2
	MFGNet[62]	70.5	82.8	79.4	65.5	79.7	81.1
	CDFM[47]	78.7	89.4	92.2	82.4	90.8	90.8
	DABNet[18]	78.7	89.4	92.2	82.4	90.8	90.8
	BABFNet [63]	**81.8**	**91.1**	**92.5**	**84.6**	**92.0**	**91.8**
Semantic Segmentation Methods	MobileNet[15]	78.2	87.0	83.5	73.0	85.0	85.2
	ICNet[64]	75.8	86.2	82.5	70.8	83.5	84.3
	DFAnet[35]	68.6	80.5	77.3	63.0	77.9	78.9
	LEDNet[65]	75.2	85.0	81.6	69.7	82.9	83.3
	BiseNet-V2[40]	80.1	90.2	**92.5**	83.4	91.4	91.3
	TopFormer[46]	79.3	89.8	91.6	82.6	90.8	90.7
	LGCNet	80.2	90.3	92.3	83.5	91.4	91.3
	+/− diff	-1.6	-0.8	-0.2	-1.1	-0.6	-0.5

which subsequently results in lower recall. In methods of cloud detection, CDFM [47] and DABNet [18] are two methods that target lightweight cloud detection. Our proposed LGCNet has 0.6% and 2.6% performance improvement over CDFM [47] and DABNet [18] in metrics MIoU. In methods of lightweight semantic segmentation, BiseNet-V2 [40] designs a subbranch for semantic extraction, and it achieves outstanding performance; however, LGCNet gains 0.3% performance increment in metric MIoU. TopFormer [46] designs a model based on transformer for mobile semantic segmentation. LGCNet has 2.9% improved performance in metrics MIoU over it.

Table III presents the experimental results from the thin-cloud sub-dataset. BABFNet [63] excels in all evaluation metrics, demonstrating exceptional detection performance. The specially designed boundary-aware module significantly contributes to the efficient detection of thin clouds. Our proposed LGCNet, although securing second place in all metrics, also performs admirably. Considering the detection accuracy metric MIoU, BiseNet-V2 [40] and TopFormer [46] are lose behind. They both contain specific modules for semantic extraction.

For visual comparison, we take five samples from the test set of the LandSat8 dataset. Samples cover plain background scenarios (sample 3) thin-cloud scenarios (samples 1, 2, and 4) and cloud-ice coexisting scenarios (sample 5). In ground truth image, white areas represent cloud and black areas represent noncloud. The red areas represent false-negative detection and the blue areas represent false-positive detection. Fig. 9 shows the visual comparison with cloud detection methods. From the results, LGCNet and BABFNet [63] perform better, especially in thin-cloud areas. In samples 2 and 4, the boundaries of thin-cloud are heavily impacted by background, and LGCNet performs much better. In samples 1 and 5, LGCNet has more false-negative detection than BABFNet [63], and it is caused by the adoption of dilate convolution in order to make the model lightweight. In addition, DABNet [18] also employs deformable convolution to capture multiscale context features. LGCNet, on the other hand, incorporates additional

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Fig. 9. Visual comparisons of different cloud detection methods in the scene of five examples from the LandSat8 dataset. White area represents cloud and black area represents noncloud. In addition, red area represents false-negative detection and blue area represents false-positive detection. (a) False-color RSI (image). (b) Ground truth. (c) Result of U-Net [58]. (d) Result of CS-CNN [59]. (e) Result of RS-Net [60]. (f) Result of MF-CNN [61]. (g) Result of CDFM [47]. (h) Result of MFGNet [62]. (i) Result of DABNet [18]. (j) Result of BABFNet [63]. (k) Result of LGCNet (ours).



Fig. 10. Visual comparisons of different lightweight semantic segmentation methods in the scene of five examples from the LandSat8 dataset. White area represents cloud and black area represents noncloud. In addition, red area represents false-negative detection and blue area represents false-positive detection. (a) False-color RSI (image). (b) Ground truth. (c) Result of MobileNet [15]. (d) Result of ICNet [64]. (e) Result of DFAnet [35]. (f) Result of LEDNet [65]. (g) Result of BiseNet-V2 [40]. (h) Result of TopFormer [46]. (i) Result of LGCNet (ours).

global context information. With this inclusion, it outperforms DABNet, especially in thin-cloud scenario (sample 2).

Fig. 10 presents the visual comparison with lightweight semantic segmentation methods. The results are similar, and our proposed LGCNet achieves outstanding detection performance, especially in complex scenarios. In TopFormer [46], a token pyramid module is utilized to encode multiscale features within the encoder. In comparison, this enhances performance by extracting additional dense multiscale features through the proposed module LSB. In complex thin-cloud scenarios, LGCNet demonstrates superior performance over TopFormer (sample 2).

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TABLE IV

COMPARISON WITH CLOUD DETECTION AND LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS ON THE MODIS DATASET. THE LAST LINE SHOWS THE DIFFERENCE BETWEEN OUR LGCNET AND THE HIGHEST VALUE OF OTHER METHODS

Category	Methods	Jaccard	Precision	Recall	MIoU	Overall	F1-Score
Cloud Detection Methods	UNet[58]	89.7	92.6	91.5	86.8	93.2	92.1
	CS-CNN [59]	87.2	91.2	90.7	83.4	91.5	82.3
	RS-Net[60]	89.7	95.2	92.5	86.6	93.3	93.8
	MF-CNN[61]	**90.0**	92.4	93.5	86.9	93.4	93.0
	MFGNet[62]	87.7	90.5	93.5	83.5	91.6	92.0
	CDFM[47]	89.9	93.2	92.7	86.9	93.4	92.9
	DABNet[18]	87.1	93.7	93.6	88.1	93.7	93.6
	BABFNet [63]	80.7	90.3	90.0	81.9	90.1	90.1
Semantic Segmentation Methods	MobileNet[15]	82.2	87.3	86.5	76.9	87.7	81.4
	ICNet[64]	85.2	87.5	88.3	79.8	89.6	87.9
	DFAnet[35]	81.3	85.9	84.6	74.5	86.5	85.3
	LEDNet[65]	85.8	89.8	88.7	80.6	90.1	89.3
	BiseNet-V2[40]	88.7	92.2	92.0	85.3	92.5	92.1
	TopFormer[46]	85.1	92.7	92.6	86.2	92.7	92.6
	LGCNet	**90.0**	**95.7**	**95.6**	**90.9**	**95.3**	**95.7**
	+/− diff	0.0	+0.5	+1.9	+2.8	+1.6	+1.9

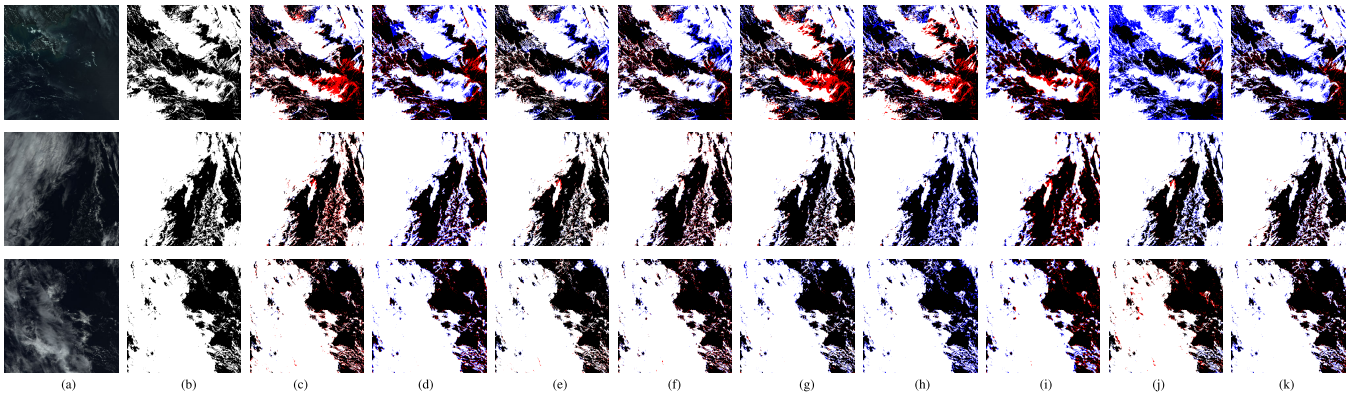


Fig. 11. Visual comparisons of different cloud detection methods in the scene of three examples from the MODIS dataset. White area represents cloud and black area represents noncloud. In addition, red area represents false-negative detection and blue area represents false-positive detection. (a) False-color RSI (image). (b) Ground truth. (c) Result of U-Net [58]. (d) Result of CS-CNN [59]. (e) Result of RS-Net [60]. (f) Result of MF-CNN [61]. (g) Result of CDFM [47]. (h) Result of MFGNet [62]. (i) Result of DABNet [18]. (j) Result of BABFNet [63]. (k) Result of LGCNet (ours).

C. Cloud Detection Results on MODIS Dataset

We summarize the experimental results of methods on the MODIS dataset in Table IV. Different from the LandSat8 dataset, there are much more special scenes such as oceans and polar ice regions in the MODIS dataset. Therefore, cloud boundary processing is much more important for cloud detection in the MODIS dataset. In comparison with the cloud detection methods, our proposed LGCNet achieves much better detection performance in metrics recall, MIoU, F1-Score, and overall accuracy. Even though RS-Net [60] gains the highest value in precision, our method LGCNet has a 2.7% increment in recall over it. The MF-CNN [61] has the best score in Jaccard, but LGCNet achieves higher performance in other metrics in comparison with it. From the results, the lightweight semantic segmentation methods perform worse than most cloud detection methods. The widely adopted dilate convolution can keep the parameter and size of the model at a low level, but detailed information is missing.

Fig. 11 shows the visual comparison of cloud detection methods with our LGCNet on three examples from the MODIS dataset. LGCNet gains more accuracy in thin-cloud detection, especially in complex scenes. BABFNet [63] incorporates a boundary prediction branch designed to manage the intricate detection in areas of cloud boundaries. There are still significantly more false negatives compared to LGCNet. From the results, the acquired adaptive and diverse contextual information in LGCNet does contribute to enhancing the performance of detection. Similar to the results seen with LandSat8, DABNet [18] demonstrates a higher rate of false negatives (samples 1 and 2), underscoring that dense context is crucial for accurate cloud detection.

In Fig. 12, the results show that LGCNet has much less miss detection and false detection over the compared semantic segmentation methods in scenes. The task of semantic segmentation on NSIs significantly differs from cloud detection, and the scenes in cloud detection are more complex. To obtain a high detection accuracy, a specific design on these scenes is required. TopFormer [46] encodes multiscale features and the global context; BiseNet-V2 [40] designs a branch for semantics encoding. However, due to the unique characteristics of thin clouds, including their transparency and discontinuity, the semantics encoded by them are insufficient. In comparison,

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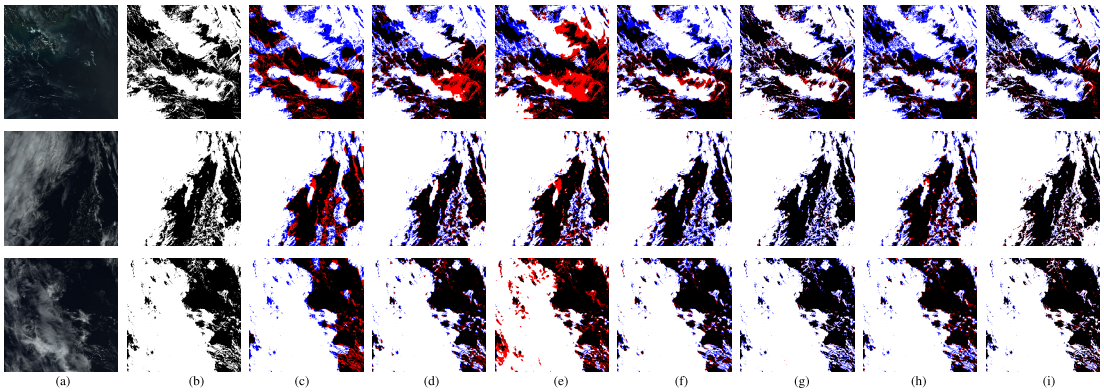


Fig. 12. Visual comparisons of different lightweight semantic segmentation methods in the scene of three examples from the MODIS dataset. White area represents cloud and black area represents noncloud. In addition, red area represents false-negative detection and blue area represents false-positive detection. (a) False-color RSI (image). (b) Ground truth. (c) Result of MobileNet [15]. (d) Result of ICNet [64]. (e) Result of DFAnet [35]. (f) Result of LEDNet [65]. (g) Result of BiseNet-V2 [40]. (h) Result of TopFormer [46]. (i) Result of LGCNet (ours).

TABLE V

EFFICIENCY COMPARISON ON THE LANDSAT8 DATASET. THE LAST LINE SHOWS THE DIFFERENCES BETWEEN OUR METHOD AND THE BEST SCORES

Category	Methods	GFLOPs	Params (10^6)	Time/image (ms)	Model size (M)	Occupied Memory (M)
Cloud Detection Methods	UNet[58]	147.50	34.53	27.5	102	1163.25
	CS-CNN [59]	11.37	8.63	**7.2**	104	97.45
	RS-Net[60]	92.40	7.85	25.0	60	498.90
	MF-CNN[61]	70.57	17.41	29.1	133	412.85
	MFGNet[62]	68.65	7.83	25.1	90	1613.01
	CDFM[47]	5.61	0.73	21.6	9.0	500.00
	DABNet[18]	10.61	4.01	26.2	48.2	155.77
	BABFNet [63]	133.4	39.6	76.2	475.5	3978.0
Semantic Segmentation Methods	MobileNet[15]	1.72	4.50	8.4	38	169.60
	ICNet[64]	1.88	4.98	20.9	58	79.57
	DFAnet[35]	1.01	2.17	43.1	26	124.45
	LEDNet[65]	3.56	0.92	30.0	11	213.17
	BiseNet-V2[40]	7.25	3.47	12.1	42	175.3
	TopFormer[46]	2.81	1.38	17.2	16.9	581.24
	LGCNet	**0.46**	**0.34**	22.3	**4.4**	**56.5**

the specific module design of LSB and GFTB in LGCNet helps effectively extract diverse semantic information, thereby enhancing detection accuracy.

D. Efficiency Evaluation

We evaluate the efficiency of compared models on the image from the LandSat8 dataset, which is of size 4 , and the MODIS dataset, which is of size 10 . The efficiency-related metrics collected on Ubuntu 16.05 with two RTX 390 GPUs are stated in Table V of LandSat8, Table VI of MODIS, and Fig. 13. As shown in the results of LandSat8, our proposed LGCNet has the fewest parameters and the least computation complexity denoted by GLOPs. In comparison with DFAnet [35], which has a GLOPs score of 1.01, our LGCNet needs 1/2 of the GLOPs and occupies less than 1/2 of its storage memory. CDFM [47] has much fewer parameters by reusing kernels, but the GLOPs are still large. Even though BABFNet [63] gains better performance in thin-cloud detection, the model requires much more resource than others. For the cloud detection task on satellite, keeping these metrics in a small amount is crucial. The limited storage and resource usage call for models of small size and need a few calculations. When considering the inference time, our LGCNet needs a longer time than most lightweight semantic segmentation methods

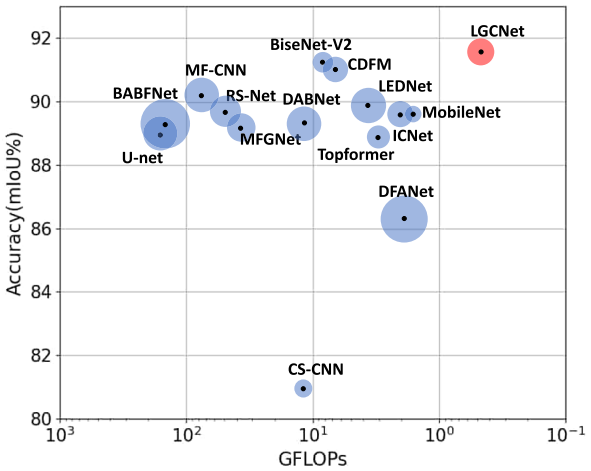


Fig. 13. GFLOPs, MIoU performance, and inference speed on the LandSat8 dataset. X-axis is the GFLOPs, Y-axis is the accuracy presented by MIoU, and the bubble size represents the inference time per image. The smaller bubble, the faster inference speed. Here, our method is compared with cloud detection models (U-Net [58], Deeplab [38], CS-CNN [59], RS-Net [60], MF-CNN [61], MFGNet [62], CDFM [47], and DABNet [18]) and lightweight semantic segmentation methods (MobileNet [15], ICNet [64], DFAnet [35], LEDNet [65], BiseNet-V2 [40], and TopFormer [46]).

to generate the cloud mask. However, most cloud detection methods require more inference time. As stated in [37], the

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TABLE VI

EFFICIENCY COMPARISON ON THE MODIS DATASET. THE LAST LINE SHOWS THE DIFFERENCES BETWEEN OUR METHOD AND THE BEST SCORES

Category	Methods	GFLOPs	Params (10^6)	Time/image (ms)	Model size (M)	Occupied Memory (M)
Cloud Detection Methods	UNet[58]	263.1	34.53	33.6	264	2068
	CS-CNN [59]	24.4	8.63	**10.4**	99	202.3
	RS-Net[60]	31.6	7.85	27.2	133	887.0
	MF-CNN[61]	125.7	17.41	27.5	272	734.0
	MFGNet[62]	23.9	7.83	30.2	63	2873.5
	CDFM[47]	9.4	0.73	32.6	9.1	872.1
	DABNet[18]	18.9	4.01	21.0	48.3	272.9
	BABFNet [63]	237.68	39.6	56.8	475.5	7584.0
Semantic Segmentation Methods	MobileNet[15]	2.47	4.50	16.7	39	391.9
	ICNet[64]	3.84	4.98	36.0	58	141.5
	DFAnet[35]	1.82	2.17	48.3	27	221.2
	LEDNet[65]	5.81	0.92	42.9	12	379.0
	BiseNet-V2[40]	13.2	3.47	14.2	42	311.8
	TopFormer[46]	6.29	1.38	20.0	17	816.4
	LGCNet	**0.84**	**0.34**	26.8	**4.5**	**101.8**

TABLE VII

EFFICIENCY COMPARISON ON LANDSAT8 AND MODIS DATASETS ON EDGE COMPUTING DEVICE NVIDIA JETSON XAVIER NX

		LandSat8			MODIS		
Category	Methods	Inference Time (ms)	Power/image (W)	Avg Power/s (W)	Inference Time (ms)	Power/image (W)	Avg Power/s (W)
Cloud Detection Methods	UNet[58]	456	9.53	20.9	-	-	-
	CS-CNN [59]	62	1.22	19.6	164.1	3.05	18.6
	RS-Net[60]	130	2.56	19.7	157.5	3.31	21
	MF-CNN[61]	230	4.76	20.7	305.7	6.45	21.1
	MFGNet[62]	213	4.47	21.0	-	-	-
	CDFM[47]	145	2.78	19.2	263	5.05	19.2
	DABNet[18]	170	2.72	16.0	345	5.93	17.2
	BABFNet [63]	-	-	-	-	-	-
Semantic Segmentation Methods	MobileNet[15]	**16**	**0.32**	20.16	**26.8**	**0.53**	19.8
	ICNet[64]	43	0.42	9.74	59.4	0.59	9.97
	DFAnet[35]	98	0.79	**8.10**	119.8	1.08	**9**
	LEDNet[65]	76	1.11	14.6	83.3	1.54	18.5
	BiseNet-V2[40]	59	1.01	17.2	71.1	1.46	20.6
	TopFormer[46]	73	0.86	11.8	79.1	1.08	13.6
	LGCNet	88	1.05	12.02	114	1.63	14.3

degree of parallelism is reduced by network fragmentation in execution. Therefore, the "multipath" structure slows the inference in devices with strong parallel computing powers such as GPU. The models CS-CNN [59] and MobileNet [15] organize convolution operations in sequential, and less time is required in mask inference. In contrast, DFAnet [35] targets reducing the parameters and calculation complexity. It adopts lightweight subnetworks and Xception [36] for the backbone. The parallel subnetworks and parallel subbranches in blocks of Xception further make the network more fragmental and slow down the mask inference. In LGCNet, the LSB consists of two parallel branches for different semantic extraction. Therefore, the inference time of our LGCNet is longer. Since the images from MODIS are on the scale of those from LandSat8, from Table VI, the models need more computation and a longer time for inference. Our proposed LGCNet occupies still the least memory in the inference phase. In conclusion, our method needs the least calculation and storage resources when compared with other methods. It is the most suitable model to be deployed on satellites.

In addition, we conduct experiments on the edge computing device: NVIDIA Jetson Xavier NX. The results are summarized in Table VII. The inference time is used for generating the cloud mask for an RSI of the size 384×384 for LandSat8 and 512×512 for MODIS. Then, we collect the average used power in the cloud masks inference phase, and the sampling time of power is 5 points/s. To calculate the power used for a single RSI inference, we multiply the used inference time and the average power per second, and the results are presented by metric: power/image in unit watt. From Table VII, the image in MODIS is bigger than that in LandSat8 and needs more resource in the inference phase. However, the results of models on LandSat8 and MODIS are consistent. The average power consumption per second of MobileNet [15] is high. However, it needs the least time in single cloud mask generation and requires the least used power per RSI in comparison with other methods. BABFNet [63] requires so much resources that it cannot run on mobile devices. When comparing all included methods, the cloud detection methods spend more time and have higher power

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TABLE VIII
PERFORMANCE OF NETWORK WITH COMBINATION OF INDIVIDUAL MODULE ON LANDSAT8

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score
WindowSize=6	87.8	94.7	96.1	91.2	96.1	95.4
WindowSize=12	88.4	95.1	96.1	91.6	96.3	95.6
WindowSize=24	86.9	94.1	96.0	90.5	95.8	95.0

TABLE IX
PERFORMANCE OF NETWORK WITH COMBINATION OF INDIVIDUAL MODULE ON MODIS

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score
WindowSize=8	89.9	95.6	95.6	90.7	95.2	95.6
WindowSize=16	90	95.7	95.6	90.9	95.3	95.7
WindowSize=32	89.8	95.5	95.6	90.6	95.1	95.5

TABLE X
ABLATION STUDY OF MODULES: LOCAL-WINDOW TRANSBLOCK AND LIGHTWEIGHT DEFORMABLE CONV IN LSB ON LANDSAT8

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
LSB (Lightweight Deformable Conv)	86.7	94.0	96.0	90.4	95.7	95.0	0.44	0.26	47.0
LSB (Local-Window TransBlock)	87.8	94.7	96.0	91.2	96.1	95.3	0.39	0.30	48.0
LSB	88.4	95.1	96.1	91.6	96.3	95.6	0.46	0.34	56.5

TABLE XI
ABLATION STUDY OF MODULES: LOCAL-WINDOW TRANSBLOCK AND LIGHTWEIGHT DEFORMABLE CONV IN LSB ON MODIS

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
LSB (Lightweight Deformable Conv)	89.6	95.7	95.3	90.5	95.1	95.5	0.79	0.26	84.9
LSB (Local-Window TransBlock)	89.4	95.5	95.3	90.3	95	95.4	0.71	0.29	86.6
LSB	90	95.7	95.6	90.9	95.3	95.7	0.84	0.34	101.8

TABLE XII
ABLATION STUDY OF LSB: CONNECTION WAY OF A LIGHTWEIGHT DEFORMABLE CONVOLUTION BLOCK AND LOCAL-WINDOW TRANSFORMER BLOCK IN LSB ON LANDSAT8

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
Sequence (Lightweight Deformable Conv + Local-Window TransBlock)	86.1	93.7	95.8	90.0	95.6	94.7	0.43	0.32	56.5
Sequence (Local-Window TransBlock + Lightweight Deformable Conv)	87.6	94.8	95.8	91.0	96.0	95.3	0.43	0.32	56.5
Parallel	88.4	95.1	96.1	91.6	96.3	95.6	0.46	0.34	56.5

consumption than most semantic segmentation methods. Since the included semantic segmentation methods are designed to be lightweight models and to fit mobile devices, efficiency is considered in model design. The inference time and power consumption of our proposed LGCNet are comparable to the lightweight semantic segmentation methods and much less than the cloud detection methods. Consequently, LGCNet can achieve a state-of-the-art tradeoff between effectiveness and efficiency.

E. Ablation Studies

To evaluate the effectiveness of our LGCNet and the proposed modules in it, we conduct ablation studies on LandSat8 datasets. These models are trained on the training set and then

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TABLE XIII
ABLATION STUDY OF LSB: CONNECTION WAY OF A LIGHTWEIGHT DEFORMABLE CONVOLUTION BLOCK AND LOCAL-WINDOW TRANSFORMER BLOCK IN LSB ON MODIS

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
Sequence (Lightweight Deformable Conv + Local-Window TransBlock)	89.2	94.8	94.8	90.1	94.8	94.8	0.79	0.32	101.8
Sequence (Local-Window TransBlock + Lightweight Deformable Conv)	89.1	94.8	94.8	90.1	94.8	94.8	0.79	0.32	101.8
Parallel	90.0	95.7	95.6	90.9	95.3	95.7	0.84	0.34	101.8

TABLE XIV
ABLATION STUDY ON THE LIGHTWEIGHT DEFORMABLE CONVOLUTION ON LANDSAT8

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
Normal Convolution	87.0	94.4	95.8	90.6	95.8	95.1	0.45	0.34	51.8
Lightweight Deformable Convolution	88.4	95.1	96.1	91.6	96.3	95.6	0.46	0.34	56.5

TABLE XV
ABLATION STUDY ON THE LIGHTWEIGHT DEFORMABLE CONVOLUTION ON MODIS

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
Normal Convolution	88.7	94.5	94.5	89.6	94.6	94.5	0.81	0.34	93.4
Lightweight Deformable Convolution	90	95.7	95.6	90.9	95.3	95.7	0.84	0.34	101.8

tested on the test set. We compare the models' performance metrics, including Jaccard index, precision, recall, MIoU, F1-Score, and overall accuracy. For efficiency comparison, we include representative metrics: GFLOPs, number of parameters, and occupied memory.

1) Window Size in Local-Window TransBlock: The local-window TransBlock is designed for local semantic extraction. By utilizing the advantages provided by the transformer, local-window TransBlock can obtain intensive attention-based semantics inside the partitioned windows. The window size determines the region of semantics. We set the size of the window with numbers 6, 12, and 24. The test results are summarized in Tables VIII and IX. From the results, when the window size equals 12, the model gains the best value in metrics MIoU in both LandSat8 and MODIS datasets. When the number is small, not enough semantics are included. When the number is large, more noise will impact the detection accuracy.

2) Local Semantic Block: The LSB is proposed for semantic extraction. The included two branches are of a lightweight deformable convolution and a proposed local-window transformer block. The lightweight deformable convolution targets capturing the clouds' semantics within the cloud boundary of various geometric structures. Also, the local-window transformer block aims to extract the intensive attention-based semantics. Both of them contribute to cloud detection performance. From results on LandSat8 stated in Table X, when the LSB only contains blocks built on the lightweight deformable convolution, the model obtains accuracy in metrics MIoU 90.4% and GFLOPs 0.44. With the local-window transformer block, the MIoU is improved to 91.2% and the calculation complexity is reduced. By including both blocks in LSB, we get the best performance with MIoU 91.6%, and there is only a slight increment in calculation complexity and occupied memory. From results on MODIS shown in Table XI, the local-window transformer block and the lightweight deformable convolution boost the detection performance by 0.4% and 0.6% on metric MIoU, respectively. Similar to the results on LandSat8 dataset, the increment on resource usage is little.

3) Connection Ways in LSB: In LSB, the lightweight deformable convolution and the local-window transformer block are designed to extract two-scale diverse semantics. They are connected in parallel to boost the feature extraction. To evaluate the way of connection, Tables XII and XIII present the detection results on LandSat8 and MODIS with the parallel connection and the sequential connection, respectively. Here, the sequential connection contains two ways: first, the local-window transformer block follows the lightweight deformable convolution, and second, the opposite. From the results, the model with parallel connection in LSB needs more calculation but gains the best detection performance based on metric MIoU both on LandSat8 and MODIS datasets.

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TABLE XVI
ABLATION STUDY OF FEB ON LANDSAT8: THE NUMBER OF INCLUDED FEM

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
FEM in FEB = 1	86.1	93.7	95.8	90.0	95.6	94.7	0.44	0.32	50.3
FEM in FEB = 2	88.4	95.1	96.1	91.6	96.3	95.6	0.46	0.34	56.5
FEM in FEB = 3	87.6	94.8	95.8	91.0	96.0	95.3	0.36	0.36	62.4

TABLE XVII
ABLATION STUDY OF FEB ON MODIS: THE NUMBER OF INCLUDED FEMS

Model	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPs	Params (10^6)	Occupied Memory (M)
FEM in FEB = 1	89.9	95.2	95.1	90.7	95.2	95.1	0.81	0.32	91.3
FEM in FEB = 2	90.0	95.7	95.6	90.9	95.3	95.7	0.84	0.34	101.8
FEM in FEB = 3	90.0	95.2	95.1	90.8	95.2	95.1	0.97	0.36	112.3

TABLE XVIII
PERFORMANCE OF NETWORK WITH COMBINATION OF INDIVIDUAL MODULE ON LANDSAT8

Backbone	LSB	GFTB	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPS	Params (10^6)	Occupied Memory (M)
✓			86.6	93.9	96.0	90.4	95.7	94.9	0.35	0.13	31.1
✓	✓		87.5	94.5	96.1	91.0	96.0	95.3	0.45	0.25	54.8
✓		✓	86.9	94.2	95.9	90.5	95.8	95.0	0.37	0.22	32.8
✓	✓	✓	88.4	95.1	96.1	91.6	96.3	95.6	0.46	0.34	56.5

TABLE XIX
PERFORMANCE OF NETWORK WITH COMBINATION OF INDIVIDUAL MODULE ON MODIS

Backbone	LSB	GFTB	Jaccard	Precision	Recall	MIoU	OA	F1-Score	GFLOPS	Params (10^6)	Occupied Memory (M)
✓			89.1	95.3	95.3	90.1	94.8	95.3	0.65	0.13	56.4
✓	✓		89.5	95.4	95.4	90.4	95	95.4	0.81	0.25	98.6
✓		✓	89.5	95.1	95.8	90.4	95	95.4	0.67	0.22	59.6
✓	✓	✓	90	95.7	95.6	90.9	95.3	95.7	0.84	0.34	101.8

4) Lightweight Deformable Convolution: The lightweight deformable convolution in LSB is designed to tackle the semantic extraction in various geometric areas of the cloud. With effective feature extraction, the model's cloud detection ability can be boosted. To evaluate the module, we built LSB in LGCNet with normal convolution and lightweight deformable convolution separately and compared the performance of each model. In Tables XIV and XV, the models' performance on LandSat8 and MODIS is stated. From the results, the detection accuracy in metric MIoU is increased by 1% and 1.3% in each dataset. The lightweight deformable convolution needs a little more computation than the normal convolution because it needs extra offset calculation. Consequentially, the semantics introduced by lightweight deformable convolution can effectively enhance the model detection accuracy with little more extra resources.

5) Number of FEM: FEB is the basic block to build the FEM and is designed to extract expressive features with low resource consumption. With less stacked FEB, FEM consumes few resources, but its feature encoding capacity decreases. With more stacked FEB, there might be an overfitting problem, and more resources are needed. In the current FEM, the proposed FEB is stacked twice for feature encoding. Tables XVI and XVII show the model detection results with the number of FEB equaling 1, 2, and 3 on LandSat8 and MODIS, respectively. From tests, the model with FEM of two FEBs has the best number on all metrics.

6) Modules in Model: In our proposed LGCNet, we design the LSB and the GFTB to capture local and global semantics. Experiments are conducted to evaluate their performance. From Table XVIII, on LandSat8, the model without LSB and GFTB only gains MIoU of score 90.4%. By including LSB in the model, MIoU is increased to 91.0% with 0.1 increments in GFLOPs. When the module GFTB is added, there is a 0.1% increment in metrics MIoU and the calculation complexity rises slightly (0.02 in GFLOPs). With the adoption of both modules, the detection accuracy is improved from 90.4% to 91.6% in MIoU. The results on MODIS are shown in

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Table XIX. Similar to LandSat8, GFTB and LSB boost the model performance on metric MIoU with 0.5% and 0.5%, respectively. Also, the calculation complexity increases by 0.03 with GFTB and 0.17 with LSB in GFLOPs. In conclusion, semantics promotes the performance of cloud detection.

VI. Conclusion and Future Work

In this work, we propose the DL-based lightweight cloud detection model LGCNet for RSIs. The designed LGCNet targets improving the cloud detection accuracy by adding expressive and diverse semantics. LGCNet is built on an encoder–decoder architecture. We design a lightweight feature encoding module to build the backbone. Besides, two modules for semantic extraction are stated. In the LSB, the semantics within the cloud boundary, which has various geometric structures and intensive semantics inside the local window, can be captured at the same time. The GFTB aims at extracting high-level global semantics for overall context information. We conduct intensive experiments on LandSat8 and MODIS datasets. The results present that LGCNet can gain state-of-the-art performance over the existing cloud detection and lightweight semantic segmentation methods. For efficiency validation, we compare the related metrics obtained in the server and the edge computing device. LGCNet has the best tradeoff between accuracy and resource usage. In future work, we will continue working on reducing the model's resource usage on the edge computing device, which will be more suitable for satellites.

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